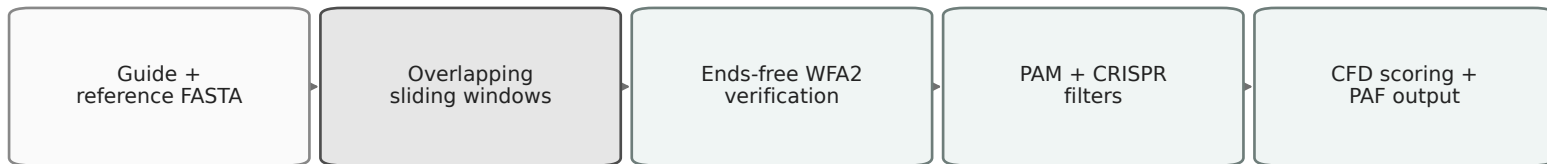
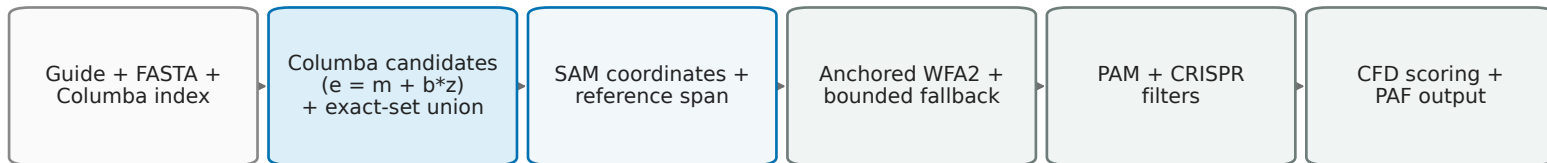


Integration workflow

Original CRISPRapido



Columba-enabled CRISPRapido



Columba replaces candidate generation; WFA2 verification, PAM/filtering, CFD scoring, and PAF reporting remain downstream.